Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

q2_age_tf_activity_and_comparison.py

This addresses the accelerated ageing question properly. Rather than checking whether the mutant's transcription factors happen to show up in the ageing data, which would be biased towards finding agreement, it scores all the transcription factors independently on the ageing data and then compares the two full sets. That way each comparison gets its own answer and neither is favoured.

- **tf_activity_age.csv:** every transcription factor scored on the ageing comparison, with its activity score and p-value, ranked. This is the independent TF list for natural ageing, and it is what the earlier approach could not produce. 37 TFs come out significant, and they are dominated by metabolic and liver-type regulators, which is a completely different set from the mutant's.
- **q1_vs_age_tf_comparison.csv:** one row per transcription factor showing its activity in the mutant, its activity in ageing, whether it is significant in each, and whether the two agree on direction. Only 5 TFs are significant in both, and 3 of those 5 point in opposite directions, which is a striking way of saying the two programmes barely overlap.
- **q1_vs_age_scatter.png:** the mutant activity on the x-axis against the ageing activity on the y-axis, one dot per TF, with the ones significant in both labelled. If the mutant were simply accelerated ageing, the dots would line up along a diagonal. They do not. They form a round cloud with a correlation of about minus 0.14, which is essentially no relationship, and that is the answer to the question.